Universität Bonn

Notes for the lecture

Combinatorics, Graphs, Matroids

held by

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TEXed by

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Corrections and improvements

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Lecture

Chapter 1

Combinatorics

[08.10.2024, Lecture 1]

Goal: Count the number of elements in some finite sets.

Lemma 1.1: Fundamental counting rules

(*i*) *For pairwise disjoint and finite sets* $U_1, \dots, U_n$*, we have*

$$\left| \bigcup_{i=1}^{n} U_i \right| = \sum_{i=1}^{n} |U_i| . \tag{1.1}$$

(*ii*) *For finite setzs* $U_1, \dots, U_n$*, we have*

$$|U_1 \times \cdots \times U_n| = \prod_{i=1}^{n} |U_i| . \tag{1.2}$$

(*iii*) *If there is a bijection between two sets* S *and* T*, then* $|S| = |T|$*.*

Notation. We call a set a n-set if it is a set with cardinality $n \in \mathbb{N}$ (where $0 \in \mathbb{N}$).

Proposition 1.2. *For two* n*-sets* A *and* B*, the number of bijections between* A *and* B *is* $n!$*.*

Proof. Let U the set of bijections from A to B. Apply induction in n:

$n = 0$: Trivial, because $0! = 1 = |U|$.

$n \mapsto n + 1$: Let $n > 0$ and let $x \in A$. For $i \in B$, let U_i be the set of all bijections $f : A \to B$ with $f(x) = i$. By induction, $|U_i| = (n-1)!$ for each $i \in B$. Then, $U = \bigcup_{i \in B} U_i$, so we get

$$|U| = \sum_{i \in B} |U_i| = n \cdot (n-1)! = n!. \tag{1.3}$$

□

Corollary 1.3. *The number of permutations of an* n*-set is* $n!$*.* □

Proposition 1.4. *The number of mappings from a* k*-set* A *to an* n*-set* B *is* n^k*.*

Proof. There is a bijection between the set of all mappings from A to B and the set $\underbrace{B \times \cdots \times B}_{k \text{ times}}$.

The number of these mappings is by Lemma 1.1.(*ii*) just $|B|^k = n^k$. □

Notation. We denote the set of all mappings from A to B by B^A.

Proposition 1.5. *The number of subsets of an* n*-set is* 2^n*.*

Proof. For an n-set A, there is a bijection between its power set and the mappings from A to $\{0,1\}$. Thus, the power set has 2^n elements. □

Notation.

(*i*) We denote the power set of a set A by 2^A.

(*ii*) The *binomial coefficient* $\binom{n}{k}$ is defined as the number of k-subsets of a n-set (for $k, n \in \mathbb{N}$).

What does *counting* mean? $\rightsquigarrow$ Possible answers:

(*i*) a direct, closed formula,

(*ii*) a sum,

(*iii*) a recursive formula

[08.10.2024, Lecture 1]
[10.10.2024, Lecture 2]

Appendix

List of definitions

List of statements

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